

SANTHOSH PRABAKAR BADHRI

Principal AI Engineer | LLM Agent & Agentic Systems Architect | RAG, Semantic Data Platforms & Enterprise AI

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PROFESSIONAL SUMMARY

Principal AI Engineer with 13.5+ years of experience designing and shipping production LLM-powered agents and enterprise AI platforms across banking, financial services, healthcare, and consulting. Specialized in multi-agent orchestration and tool-calling (LangChain, LangGraph, MCP) over structured data platforms on Snowflake, AWS, and Microsoft Azure — building semantic layers, NL-to-SQL pipelines, and verified query grounding (Snowflake Cortex Analyst, Cortex Search) that let business users query complex transactional and financial data through natural language. Deep expertise in Python-based RAG and document intelligence pipelines (chunking, embedding, hybrid retrieval, cross-encoder reranking) for summarisation, content generation, and business-document workflows, backed by production-grade agent evaluation, hallucination mitigation, and guardrail frameworks (LangSmith, Ragas, DeepEval, Langfuse). Comfortable owning the full lifecycle from scoping use cases with business SMEs through cloud deployment (AWS Bedrock, Microsoft Azure, SageMaker-adjacent MLOps, Lambda, S3) in sprint-sequenced delivery, and connecting Python and React-based agent surfaces to the systems where end users actually work.

CORE COMPETENCIES

- LLM Agent Development:** Multi-agent systems, tool-calling, chain-of-thought reasoning, structured outputs, context management — Claude, GPT-4, Gemini, Llama2, Amazon Bedrock
- Agentic Orchestration:** LangChain, LangGraph, Model Context Protocol (MCP) server/client design, agent graphs, routing, disambiguation, fallback handling, multi-step task decomposition
- RAG & Document Intelligence:** Advanced RAG, semantic & hybrid search, BM25, dense/sparse retrieval, document chunking, embeddings, cross-encoder reranking, summarisation & content generation
- Data Platforms & Semantic Modelling:** Snowflake Cortex Analyst & Cortex Search, semantic views, NL-to-SQL pipelines, complex SQL joins/CTEs, Databricks Vector Search, star-schema data modelling
- Agent Evaluation & Guardrails:** LangSmith, Ragas, DeepEval, Phoenix, Helicone, hallucination mitigation, output validation, human-in-the-loop review, ABAC/RBAC-based safety guardrails
- Cloud & Deployment:** AWS (Bedrock, EC2, S3, Lambda, IAM, ECS, EKS, OpenSearch), Microsoft Azure (MS Azure), Docker, Kubernetes, MLflow, CI/CD & GitHub Actions
- Enterprise Integration:** Connecting Python and React-based agents to Jira, internal data portals, SSO/enterprise auth, cross-functional reporting surfaces, stakeholder-facing dashboards

TECHNICAL SKILLS

LLM / Agent APIs	Claude API, OpenAI API, Amazon Bedrock, Gemini, Llama2, Ollama, CrewAI (foundational), Semantic Kernel (working knowledge)
Orchestration	LangChain, LangGraph, Model Context Protocol (MCP), AI Agent Orchestration, Prompt Engineering, Structured Outputs, Function/Tool Calling
RAG & Retrieval	RAG, Semantic Search, Hybrid Search, BM25, Dense/Sparse Retrieval, Embeddings, Cross/Bi-Encoders, Reranking Models
Data & Semantic Layer	Snowflake Cortex Analyst, Cortex Search, Databricks Vector Search, Pinecone, ChromaDB, Qdrant, FAISS, SQL (complex joins, CTEs), MySQL
Backend / Services	Python (async), FastAPI, Pydantic, Java, PHP, REST API design, tool-interface & data-pipeline engineering
Evaluation & Observability	LangSmith, Ragas, DeepEval, Langfuse, Phoenix, Helicone, OpenTelemetry, AI Tracing, Token Usage Analytics
Cloud & Infrastructure	AWS (EC2, IAM, S3, Lambda, ECS, EKS, Bedrock), Microsoft Azure (MS Azure), OpenSearch, Docker, Kubernetes, CI/CD & GitHub Actions, MLflow
Frontend & Full-Stack	Python, React (16, 18), TypeScript, JavaScript, Vue2, HTML5, CSS3, Bootstrap, Styled Components, SantyCSS, Component Design Systems

PROFESSIONAL EXPERIENCE

Clearintelligence — Lead AI Consultant & Enterprise Architect

Jul 2025 - Jul 2026

- **Multi-Agent LLM Platform:** Architected and delivered a production conversational AI platform on Snowflake Cortex Analyst, using Python-based tool-calling and structured-output workflows to translate natural-language business questions into verified, semantically grounded SQL over financial and operational data.
- **Semantic Layer & NL-to-SQL:** Designed a dual semantic data-view layer (company-month and cost-bucket views) that grounds LLM agents in structured Snowflake data, enabling accurate NL-to-SQL query generation without hallucinated joins or metrics.
- **RAG & Document Intelligence:** Built enterprise-scale RAG pipelines — chunking, embedding, hybrid/semantic search, cross-encoder reranking — over unstructured investment PDFs via Databricks, improving grounding accuracy for document-derived answers.
- **Agent Evaluation & Guardrails:** Established evaluation and observability pipelines (prompt testing, retrieval accuracy validation, hallucination mitigation, latency/token telemetry) to continuously monitor and improve production agent reliability.
- **Cloud Deployment:** Deployed AI services on AWS/Bedrock and Microsoft Azure integration patterns using Python (FastAPI) and React-based front-end dashboards, with streaming responses and MLOps pipelines (MLflow, Docker, Kubernetes), cutting model delivery lifecycle time by 25%.
- **AI Transformation Strategy:** Orchestrated wave-sequenced delivery of enterprise AI initiatives, scoping use cases with business SMEs and standardizing sprint playbooks that cut on-time delivery risk by 35%.
- **Local Agentic AI:** Engineered a localized AI marketing agent using Ollama with custom Python tool orchestration for secure, on-prem agent workflows.
- **Mentorship:** Mentored a distributed team of 8 engineers on agentic architecture and MLOps paradigms, boosting delivery reliability by 10%.

HSBC (Purview Services) — Lead & Principal AI Engineer

Sep 2023 – Jul 2025

- **MCP-Based Multi-Agent System:** Architected a secure on-prem enterprise chatbot on Llama2, LangChain, and LangGraph, and innovated a custom Python-based Model Context Protocol (MCP) interface connecting the LLM directly to the Jira API — enabling project managers to run conversational analysis on resource allocation, timeline drift, and cost projections.
- **Enterprise RAG & Retrieval:** Built enterprise-grade RAG pipelines with document chunking, embedding generation, hybrid/dense/sparse retrieval, and cross-encoder reranking over sensitive banking documentation.
- **Agent Evaluation & Guardrails:** Developed evaluation pipelines for prompt testing, hallucination detection, and grounding verification; implemented ABAC/RBAC authorization, content filtering, and role-aware retrieval integrated with enterprise SSO.
- **Enterprise Integration:** Connected the Python-based agent platform to Jira and internal data portals; directed a React 16→18 modernization of the core internal data portal, improving load speed by 50%.
- **DevOps & Cloud:** Spearheaded a CI/CD pipeline on AWS and Microsoft Azure with high-availability load balancing and strict IAM policies, reducing release defects by 20% with zero downtime.
- **Leadership:** Directed an 8-member engineering team and managed executive stakeholder communication on platform capability roadmaps.

SystemSoft — Full Stack Developer

Apr 2023 – Aug 2023

- **AI Foundation Services:** Designed AWS-native and Microsoft Azure-ready, AI-ready backend services (Bedrock, S3, Lambda, REST APIs) to establish reusable foundations for future RAG and agentic initiatives.
- **Data & Backend Engineering:** Led a squad of 8 engineers building a high-performance Java and Python backend, React-based front-end modules, and MySQL data models, cutting backend latency by 35%.
- **Cloud Infrastructure:** Configured AWS and Microsoft Azure load balancing and S3-based CDN delivery for a secure, scalable production environment.

Callaway Golf — Lead

Oct 2022 – Jan 2023

- **eCommerce & Data Integration:** Led frontend optimization of SAP Commerce Cloud (Hybris) using React components, and maintained Python-driven REST API data connections between product catalogs and marketing analytics platforms, improving page load speed by 35%.

Infosys (Nithyo Infotech) — Senior Developer

Jul 2022 – Oct 2022

- **Reusable Platform Architecture:** Architected reusable modules for an enterprise page-builder platform (React, TypeScript, Python-based tooling) for major industrial accounts, raising automated test coverage above 90%.

10X Incubator (Cila Labs) — UI Lead

Dec 2021 – Mar 2022

- **Cloud-Native AI-Ready Deployment:** Built containerized applications on AWS (Docker, Kubernetes, S3, Bedrock integration patterns), establishing deployment architectures for AI-powered services.

Indegene — Web Lead

May 2021 – Nov 2021

- **Regulated Data Systems:** Led SEO-optimized, regulation-compliant web application delivery for healthcare/pharma clients, improving organic visibility by 20%.

Skwad — Senior Developer

Oct 2015 – Jan 2021

- **Data & Integration Engineering:** Built full-stack data-handling systems (React, PHP) and automated enterprise workflows via UiPath RPA, saving 200+ engineering hours monthly; orchestrated AWS ↔ Azure cloud migrations with IAM-based access control.

Sitosys e-Media Pvt. Ltd — Junior Web Designer

Jul 2012 – Oct 2015

- **Foundations:** Built responsive, SEO-optimized web platforms and custom CMS solutions, establishing the design-to-code discipline that underpins current UI/agent-surface work.

MAJOR AI PROJECTS

PE Insights – Conversational Enterprise AI on a Semantic Data Platform

- **Snowflake Cortex Analyst · Cortex Search · Databricks RAG · Python · FastAPI · React · Microsoft Azure** — Built a production conversational AI platform that translates natural-language questions into secure, verified SQL over financial databases via a dual semantic-view layer, combined with a Databricks RAG pipeline chunking and embedding unstructured investment PDFs for instant cross-document querying.

Enterprise On-Prem Knowledge Agent & Multi-Agent MCP

- **Llama2 · Ollama · LangChain · LangGraph · MCP · Python · Jira API · React · Microsoft Azure** — Engineered a secure on-prem multi-agent system implementing the Model Context Protocol to pull live project data from Jira, running automated forecasting on project costs and timeline delays.

Local AI Marketing Engine

- **Ollama · Local LLMs · Python · FastAPI · Microsoft Azure** — Zero-cloud-dependency agent automating content curation, copy generation, and keyword classification, eliminating third-party API spend for small-business use cases.

LEADERSHIP EXPERIENCE & OPEN SOURCE

- Spearheaded AI and Application Development practices across Cleartelligence and HSBC, reducing structural engineering debt by 30% through modern Agile and MLOps playbooks.
- Managed and mentored cross-functional engineering teams of up to 8 (data engineers, frontend developers, QA analysts) on MLOps, LLMops, and cloud security architectures across AWS and Microsoft Azure.
- Trusted technical advisor to C-suite executives and product owners — technical discovery workshops, system design translation, and ROI evaluations.
- SantyCSS (Creator & Core Maintainer): open-sourced utility-first CSS framework on npm with VS Code IntelliSense extension, Figma plugin, and PostCSS/Vite integrations.

ACHIEVEMENTS

- Zero-Outage AI Migration: led end-to-end AWS ↔ Azure platform migration with zero operational outages or security exceptions.
- Open Source Adoption: SantyCSS styling plugins increased multi-platform developer UI velocity by 40%.
- UNO No Mercy Logic Engine: real-time multiplayer card game engine with custom state synchronization, achieving under 50ms latency over WebSocket.

EDUCATION

- Bachelor of Technology (B.Tech) – Nexus College of Science & Technology, 2012
- Diploma in Animation & Interaction Design – ZICA, 2014